Paleomagnetic Results From Igneous and Sedimentary Rocks From the Michigan Basin Borehole

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The results of thermal and alternating field demagnetization of 13 specimens from a fully oriented mafic igneous core taken from a depth of 5.32 km near the center of the Michigan basin give a pole position at 9°S, 159°E ($k = 69$, $\alpha_{95} = 5°$). This pole is close to late Hadrynian and Cambrian pole positions from the North American craton. Furthermore, one specimen was cut from each of three vertical red bed core segments, unoriented in declination, from depths of 4.59, 4.09, and 4.08 km. Thermal demagnetization revealed a large secondary magnetization in these samples, which was fully eliminated at 480°C. Further thermal demagnetization revealed closely grouped characteristic inclinations that averaged 8°. By assuming the secondary magnetizations to be parallel to the present-day geomagnetic field direction it was possible to obtain characteristic declinations as well. The pole from the red beds is located at 7°N, 174°E and agrees well with the pole obtained from the late Keweenawan sediments (Freda sandstone and Nonesuch shale) of the Upper Peninsula in Michigan. These results suggest that the red beds acquired their characteristic magnetization before the igneous basement rock. If the igneous rock is an extrusive body of Keweenawan age, its characteristic magnetization must be entirely secondary.

INTRODUCTION

As part of the geophysical and geological studies associated with the Michigan basin borehole project (McClure-Sparks 1-8 well in Ithaca, Gratiot County, Michigan), paleomagnetic studies were undertaken on red bed and medium- to coarse-grained mafic igneous rock samples. The red beds are found as a 1500-m-thick and relatively undisturbed sequence below the lowest identifiable Cambrian strata of the Michigan basin; a Late Precambrian age is assumed, but the upper part could be as young as Early Cambrian. Within the red bed sequence of the borehole, igneous bodies of mafic composition were encountered at two levels; gravity studies within the borehole [*Bradley and Hinze*, 1976; *Bradley et al.*, 1976] indicate that more igneous bodies may exist at other depths. The borehole bottomed at 5.32-km depth in the lower igneous unit, from which our samples were taken. The three red bed samples were obtained to complement the paleomagnetic studies of the igneous core; these samples were drilled from the shallower depths of 4.59, 4.09, and 4.08 km.

The goals of this study were to find out whether paleomagnetism can assist in dating the sequence and to study the effects of burial temperatures on the magnetization. The experimental results will be presented here, and the age implications will be discussed. The effects of the burial temperatures have been discussed elsewhere [*Van der Voo et al.*, 1978]. The igneous core was oriented azimuthally in situ at a depth of 5.32 km by a camera-clock-compass-level bubble system attached to knives scribing the core. Thirteen specimens, distributed along at least 5 m of the core, were measured and used for statistical analysis and pole position determination. Nevertheless, it should be realized that statistically, the borehole paleomagnetic results cannot be equated with results from surface outcrops, where it is usually possible to obtain a larger number of individually oriented samples from several sites.

MEASUREMENT AND DEMAGNETIZATION

The natural remanent magnetization (NRM) of the 13 igneous specimens was measured with a Schonstedt spinner magnetometer and a ScT cryogenic magnetometer. Intensities of the NRM ranged between 0.7×10^{-2} and 1.4×10^{-2} A m^{-1} (0.7×10^{-5} and 1.4×10^{-5} emu/cm^3).

The directions of the NRM's were almost vertically downward and closely grouped. However, alternating field (af) and thermal demagnetizations removed a large and relatively 'soft' component of magnetization, which we assume to be a viscous remanence of recent origin. Above fields of 80 mT (800 Oe) or temperatures of about 450°C, another component of magnetization is revealed, with east to ESE declinations and inclinations averaging about +40°. Some decay curves and equal-area projections of the paths of the magnetization vector during progressive af or thermal demagnetization are shown in Figure 1. Orthogonal projections of the demagnetization trajectories [*Zijderveld*, 1967] are shown in Figure 2. It can be seen in the results of the af demagnetizations of Figures 1 and 2 that the intensities of magnetization at 80 mT had generally fallen to 15% or less of the intensity of the total NRM and that the samples began to show some directional instability due to anhysteretic effects associated with the af demagnetization procedure. Nevertheless, the high-coercivity components decayed approximately toward the origin above 80 mT (Figure 2), an indication that the viscous magnetization had been eliminated. Following established usage we call these components the characteristic magnetizations.

Thermal demagnetizations were carried out to complement the af studies. Good agreement was obtained between af and thermal demagnetization results (Figures 1 and 2). The viscous magnetizations appeared to be eliminated at about 450°C, 30% or less of the initial NRM intensity remaining in the samples. Above 450°C, however, further decay toward the origin was not very stable or linear. This instability was accompanied by large increases in susceptibility, an indication that mineralogical changes in the samples were responsible.

The characteristic directions selected from af demagnetization results were combined with the three groups of directions of the thermal demagnetizations at three temperatures, 400°, 450°, and 500°C, respectively, to evaluate the effects of the instability during thermal treatment. It was found that the mean directions obtained from the characteristic af directions of five specimens and the thermally obtained directions of the other eight specimens did not vary more than 5° in this inter-

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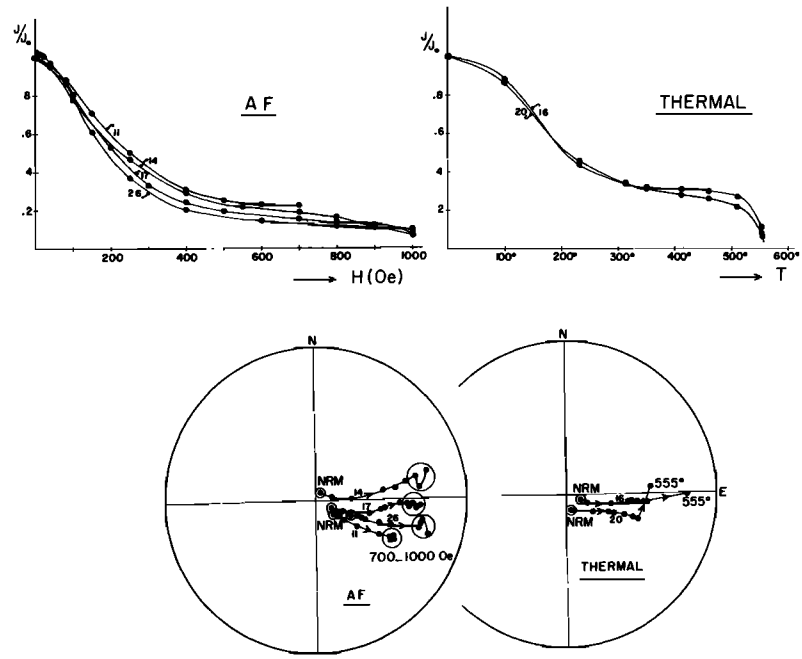


Fig. 1. (Top) Normalized intensity curves for the af and thermal demagnetizations and (bottom) equal-area projections showing the successive endpoint directions during progressive demagnetization. Samples were taken from the basic igneous rock unit at a 5.32-km depth. Closed symbols represent projections on the lower hemisphere.

val and that the directions at 450°C showed minimum dispersion. These directions have been plotted in Figure 3. On the assumption that they are of reversed polarity, the resulting North Pole position is located at 8.8°S, 158.5°E, as calculated from the mean direction with a declination of 102.3° and an inclination of +43.6° (N = 13, k = 69, and α_{95} = 5.0°).

The three red bed samples had intensities of NRM ranging from 3×10^{-2} to 6×10^{-2} A m^{-1}; they had higher NRM intensities (but lower susceptibilities) than the igneous samples. Since these samples were not azimuthally oriented, only their inclinations can be observed. However, thermal demagnetizations revealed secondary as well as characteristic magnetizations in the red bed samples; by assuming the secondary magnetizations to be parallel to the present-day in situ geomagnetic field direction it was possible to infer characteristic magnetic declinations as well. Orthogonal demagnetization diagrams, thus azimuthally oriented, are shown in Figure 4. It can be seen that the high-temperature directions of the two samples shown are mutually reversed, with approximately antipodal declinations and inclinations of −6° and +7°, respectively. The pole position calculated from these directions is located at 7.1°N, 173.8°E.

The bulk of the secondary magnetization of the red beds, with steeply downward inclinations, is eliminated below 118°C, and at about 480°C, only the characteristic magnetization remains in the samples (Figure 4). In view of the

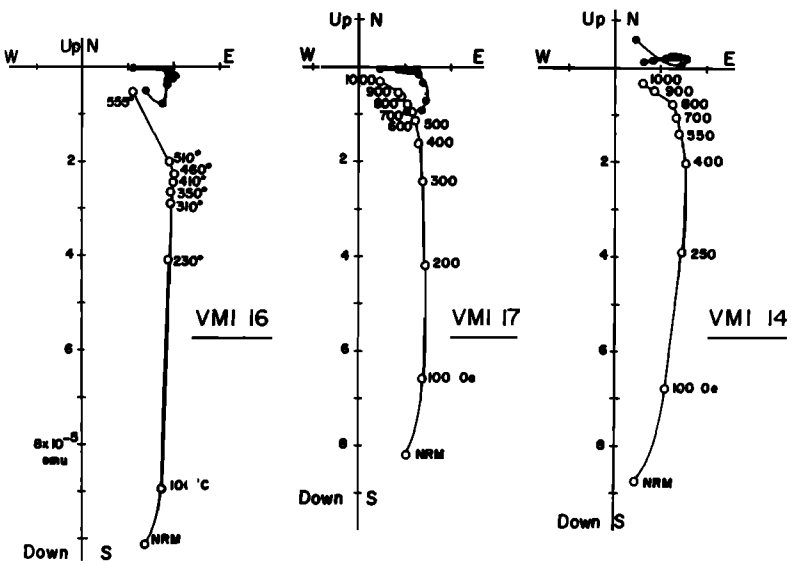


Fig. 2. Orthogonal projections [*Zijderveld*, 1967] of the successive endpoints of the magnetization vector during progressive demagnetization. Demagnetization temperatures (in degrees Celsius) and alternating field values (in oersteds; 1 Oe = 0.1 mT) are indicated. Solid symbols represent projections on the horizontal plane, and open symbols represent projections on the east-west vertical plane. Units indicated for the axes are in 10^{-3} A m^{-1} (10^{-6} emu/cm^3). Fully oriented samples were taken from the basic igneous rock unit at a 5.32-km depth.

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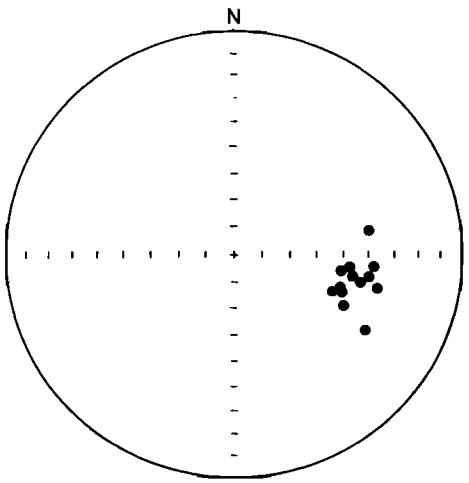


Fig. 3. Equal-area projection of the selected directions of magnetization for the 13 igneous samples.

present-day in situ temperatures of 110°–130°C [*Pollack and Watts*, 1976; *Brewster and Pollack*, 1976] it appears certain that the secondary magnetization is of recent origin and that the assumptions made to infer the declinations are warranted. Moreover, the intersection of the pre-Late-Cambrian path of apparent polar wander [*Irving and McGlynn*, 1976] with the small circle band containing all poles that would give inclinations of −6° to −10° (normal polarity) as observed at the sampling site yields a pole position very similar to the one given above.

Discussion

Since our main goal in studying the paleomagnetism of the borehole samples was to assist in dating the rocks involved, it is important to ascertain whether the characteristic magnetizations are original, i.e., whether they were acquired when the rocks were formed. For the red beds we see no reason to question the assumption that the magnetization is the initial one, acquired during or shorly after deposition. Their magnetization, which is carried by hematite, has a very high stability and blocking temperatures of up to 675°C or more. The resulting pole position, as discussed below, agrees well with the age expected from the regional tectonic setting of the red beds in the Michigan basin. The igneous rock, on the other hand, has undergone hydrothermal alteration or even metamorphism to greenschist facies according to E. W. Heinrich (personal communication, 1977), and complete remagnetization cannot a priori be excluded. The red beds do not show any magnetization which is similar to the magnetizations of the igneous core other than the steeply inclined components attributed to remagnetization in the present-day field. Therefore complete remagnetization, if it occurred, appears to have been restricted to the igneous rocks.

The pole positions of the igneous and red bed samples are plotted in Figure 5, together with published Late Precambrian and Cambrian poles for North America [*Irving and McGlynn*, 1976; *Henry et al.*, 1977; *French et al.*, 1977]. These pole positions fall into two groups. The older group, with ages between about 1100 and 1000 m.y., forms the Keweenawan trend (Table 1; Figure 5, circles). The other group, with ages between about 700 and 500 m.y., forms the late Hadrynian–Cambrian trend (Table 1; Figure 5, squares). The intervening period between 1000 m.y. and 700 m.y. has yielded to date only a limited number of poles (not plotted). These fall mostly in the general area between the above mentioned groups [*Morris*, 1977], though *Morris and Roy* [1977] recently advocated a Hadrynian age for some poles that fall close to the present-day pole. Synorogenic poles from the Grenville province have not been plotted, since its position with respect to cratonic North America is controversial [*Buchan and Dunlop*, 1976]. Moreover, they do not contribute to the discussion about the

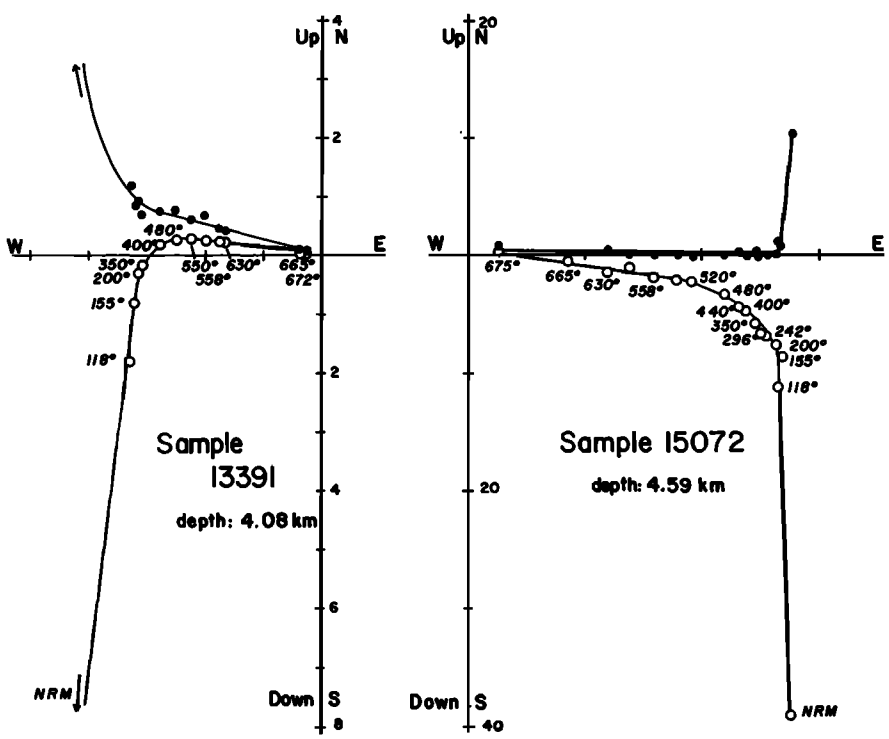


Fig. 4. Thermal demagnetization diagrams (as in Figure 2) of two red bed samples. The samples have been azimuthally oriented by assuming the secondary magnetizations to be parallel to the present-day geomagnetic field direction. Units on either axis are in 10^{-3} A m^{-1}.

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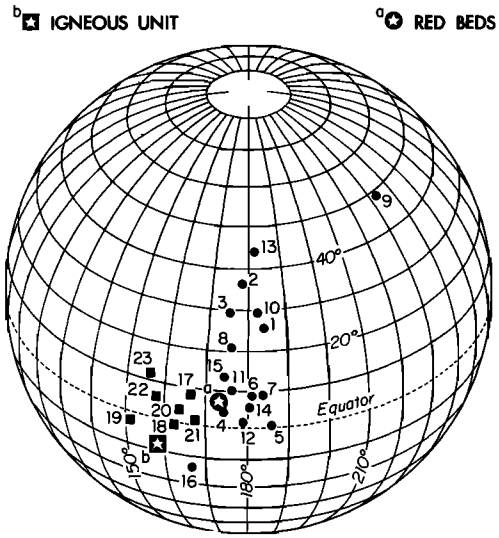


Fig. 5. Plot of Keweenawan (circles, 1100–1000 m.y.) and late Hadrynian–Cambrian (squares, 700–500 m.y.) pole positions for North America and of the poles from this study. The numbers correspond to those in Table 1.

ages of the Michigan borehole rocks, since the Grenville poles are very different and since the Michigan borehole rocks are relatively undisturbed and do not belong to the Grenville province. It is, on the other hand, quite reasonable to assume that the rocks penetrated in the Michigan borehole are analogous to the Keweenawan suite as exposed in the Lake Superior region. This assumption is supported by the similarity between the pole position of the Michigan borehole red beds and the poles of the late Keweenawan red beds (Nonesuch shale and Freda sandstone, poles 11 and 12 in Figure 5). The pole of the igneous rock of the borehole, however, is different from the middle and late Keweenawan poles of the Lake Superior region and falls near the late Hadrynian–Cambrian poles (Figure 5, squares). An interpretation of the petrography of the rock (E. W. Heinrich, personal communication, 1977) suggests that the igneous body at the base of the borehole may be an altered lava flow, with Keweenawan affinities. If this is the case, the characteristic magnetization is secondary and has replaced any initial magnetization. The mechanism by which the igneous body could have acquired the secondary magnetization is difficult to explain, as the overlying red beds, 700 m above, were not affected by late Hadrynian–Cambrian remagnetization. The thermal demagnetization curves for the igneous rock (Figure 1) show the highest blocking temperatures to be near 560°C for the carriers of the characteristic magnetization, suggesting that magnetite is the dominant carrier. A regional thermal event of sufficient intensity to remagnetize the magnetite would have very likely remagnetized the hematite in the red beds overlying the igneous body to a large extent. A possibility which cannot be excluded is that the characteristic direction of the igneous body (Figure 2) is not one of a single remanence vector but rather the resultant of the decay of two vectors of equal coercivity spectra.

In summary, our studies indicate that a late Keweenawan age (about 1000 m.y.) for the Michigan basin red beds is not unreasonable. The magnetization of the basic igneous rock is likely to be entirely secondary and may have been acquired by the unit during late Hadrynian or possibly Cambrian time. The remagnetization of the igneous body is not necessarily evidence for a thermal event, as the overlying red beds do not show any remagnetization at this time; tentatively, we suggest that the remagnetization is of chemical origin. No paleomagnetic evidence has been found for post-Cambrian thermal

TABLE 1. Paleomagnetic Pole Positions

	Approximate Age, m.y.	Formation	Pole Position	Number in Reference
b	?	Borehole igneous rock	8.8°S, 158.5°E	(this study)
a	1000?	Borehole red beds	7.1°N, 173.8°E	(this study)
23	700–500	Colorado intrusives	12.0°N, 156.0°E	7*
22	540	Wilberns	6.4°N, 158.8°E	6*
21	545	Lamotte	0.5°N, 168.0°E	5*
20	525	Wichita granites	4.0°N, 164.0°E	4*
19	700–600	Grenville dikes	3.0°S, 151.0°E	3*
18	647	Coronation sills	1.0°S, 163.0°E	2*
17	675	Franklin diabases	8.0°N, 167.0°E	1*
16	995	Upper Nankoweap	14.2°S, 166.9°E	15†
15	1005	Ferruginous Nankoweap	12.5°N, 174.4°E	14†
14	1030	Pikes Peak	6.0°N, 181.0°E	1030‡
13	1036	Nemegosenda	43.0°N, 182.0°E	1036‡
12	1050	Freda sandstone	1.2°N, 179.5°E	1050§
11	1060	Nonesuch shale	10.3°N, 176.5°E	1060§
10	1070	Mamainse, normal	29.5°N, 182.7°E	1070‡
9	1076	Mamainse, reversed	48.5°N, 232.0°E	1076‡
8	1080	Copper Harbor	21.0°N, 176.0°E	1080‡
7	1085	Sandstone, Cardenas	8.6°N, 183.7°E	12†
6	1090	Cardenas lava flows	8.3°N, 181.5°E	11†
5	1090	Cardenas lavas	0.4°N, 185.4°E	1090‡
4	1092	Upper Dox sandstone	4.6°N, 174.4°E	10†
3	1094	Upper-middle Dox	29.2°N, 175.3°E	9†
2	1096	Middle Dox sandstone	37.1°N, 178.3°E	8†
1	1100	Portage Lake lavas	25.2°N, 184.0°E	1100‡

*French et al. [1977].
†Studies by D. P. Elston and G. R. Scott, quoted in *Irving and McGlynn* [1976].
‡*Irving and McGlynn* [1976].
§*Henry et al.* [1977].

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events, which would have left a magnetic overprinting in the borehole rocks.

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